Large Project Report

Diabetes Risk Analysis and Prediction Using Apache Spark

Group Members: Blessing Taranhike Leona Chazanga

GitHub Repository: https://github.com/bvtaranh-cell/Big-Data-Analytics-

Abstract

Diabetes is a major public health concern and understanding the factors associated with the disease can help identify patterns that may be useful in assessing diabetes risk. This project used the CDC Diabetes Health Indicators dataset and Apache Spark to explore factors associated with diabetes, develop a machine learning model and evaluate Spark's performance under different executor configurations.

The original dataset contained 253,680 records. During preprocessing, duplicate records were removed, leaving 229,474 records for analysis. Descriptive statistical analysis and correlation analysis were performed to examine the relationship between diabetes status and different demographic, behavioral and health-related characteristics. The findings showed that general health, high blood pressure, difficulty walking and body mass index had some of the strongest positive correlations with diabetes status.

A logistic regression model was also developed using Spark MLlib to predict diabetes status. The model recorded an AUC of 0.809 and an accuracy of 0.852. However, its recall for the diabetes group was relatively low, indicating that it did not identify many of the individuals who had diabetes.

Finally, we evaluated Spark's performance by comparing selected workloads using 1, 2, and 4 executors. The results showed that execution times increased as the number of executors increased. This suggests that adding more executors did not improve performance in the computing environment used for this project. Overall, the project provided an opportunity to apply big data analytics and machine learning techniques to a public health dataset while examining the performance of Apache Spark.

1.Introduction

Diabetes is a major chronic disease and an important public health concern because it is associated with serious health complications and mortality (Sun et al., 2022). Several demographic, behavioral, and health-related factors, including obesity, high blood pressure, high cholesterol, physical inactivity, age and cardiovascular conditions, have been linked to diabetes (Dendup et al., 2018). Understanding these factors is important because it can help identify

patterns associated with the disease and support early identification of individuals who may be at risk.

Advances in big data analytics and machine learning have created opportunities to analyze large healthcare datasets and identify patterns related to chronic diseases. Previous studies have demonstrated the use of machine learning to predict diabetes and cardiovascular disease using demographic and clinical information (Dinh et al., 2019). However, analyzing large datasets can be computationally demanding, making efficient data-processing tools increasingly important.

Apache Spark is a distributed computing framework that supports large-scale data processing, statistical analysis and machine learning (Zaharia et al., 2016). Its capabilities make it useful for analyzing population health data and developing predictive models. This project applies these capabilities to the CDC Diabetes Health Indicators dataset to explore factors associated with diabetes and examine the performance of Spark under different executor configurations.

2.Problem Statement

Large health surveillance datasets contain valuable information that can be used to identify disease risk factors and develop predictive models. Traditional single-machine data analysis approaches may become inefficient as dataset size and analytical complexity increase. Distributed computing frameworks such as Apache Spark provide tools for processing large datasets across multiple computing resources.This project addresses two related problems. First, it investigates which health and lifestyle characteristics in the CDC Diabetes Health Indicators dataset are associated with diabetes. Second, it evaluates whether diabetes status can be predicted from these characteristics using logistic regression implemented with Spark MLlib.

3.Objectives

The project aims to:

1. Identify and analyze the demographic, behavioral and health-related characteristics most strongly associated with diabetes using the CDC Diabetes Health Indicators dataset and Spark-based statistical analysis.
2. Develop and evaluate a diabetes classification model using Spark MLlib logistic regression based on demographic, lifestyle, and health-related characteristics.
3. Evaluate the effect of Spark executor scaling by comparing the execution time of correlation analysis and logistic regression using 1, 2, and 4 executors.
4. Compare the scalability and performance of the statistical analysis and machine-learning workloads to determine whether the two methods demonstrate similar improvements as the number of Spark executors increases.

4. Team Contributions

The project was completed by two group members, with responsibilities divided as follows:

- **Blessing Taranhike:** Data preprocessing and descriptive statistical analysis.
- **Leona Chazanga:** Machine learning using Apache Spark MLlib and Spark executor performance benchmarking
- **Both members:** Comparison of results.

5.Dataset

This project used the CDC Diabetes Health Indicators dataset obtained from the UCI Machine Learning Repository. The dataset is derived from the Behavioral Risk Factor Surveillance System (BRFSS), which collects health-related information from adults in the United States.The original dataset contained 253,680 records and 21 predictor variables covering demographic, behavioral and health-related characteristics.The target variable was Diabetes_binary, which represents diabetes status:

- **0:** No diabetes
- **1:** Diabetes

Some of the variables included body mass index (BMI), high blood pressure, high cholesterol, smoking, physical activity, general health, age, income and history of heart disease.During preprocessing, duplicate records were removed, leaving 229,474 records for the main analysis.

Dataset source:
UCI Machine Learning Repository, CDC Diabetes Health Indicators
https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators

5. Methodology and Code Implementation

5.1 Data Loading and Spark Setup

The CDC Diabetes Health Indicators dataset was obtained from the UCI Machine Learning Repository using Python libraries, including PySpark, Pandas, NumPy and Matplotlib. The predictor variables (X) and target variable (y) were combined into a Pandas DataFrame and converted into a Spark DataFrame using spark.createDataFrame().A Spark session was initialized using SparkSession.builder with master("local[*]"), allowing Spark to use the available local processing threads. The driver memory was configured to 4 GB. Apache Spark was used for data preprocessing, statistical analysis, correlation analysis and machine learning.

5.2 Data Preprocessing

Data preprocessing involved examining the dataset schema, checking for missing values, removing duplicates, and converting columns into numeric double data types. No missing values were identified.Duplicate records were removed using Spark's dropDuplicates() function, reducing the dataset from 253,680 to 229,474 records. The distribution of the target variable, Diabetes-binary, was then examined, where 0 represents no diabetes and 1 represents diabetes.

Table 1. Distribution of Diabetes Categories After Preprocessing

Diabetes Category	Number Of Records(Percentage)

No Diabetes(0)	194,377(84.71)
Diabetes(1)	35,097(15.29)
Total	229,474(100)

5.3 Descriptive Statistical Analysis

Descriptive statistical analysis was performed using Spark aggregation functions to examine the characteristics of the dataset. Measures such as mean, standard deviation, minimum and maximum values were calculated for selected variables, including BMI, age category, general health, mental health and physical health.The analysis also compared the mean BMI and age category of individuals with and without diabetes to identify differences between the two groups.

Table 2. Descriptive Statistics of Selected Variables

Variable	Mean	Standard Deviation	Minimum	Maximum
BMI	28.69	6.79	12	98
Age Category	8.09	3.09	1	13
General Health	2.60	1.06	1	5
Mental Health	3.51	7.72	0	30
Physical Health	4.68	9.05	0	30

The results show that the average BMI was 28.69, while the average age category was 8.09. General health had a mean of 2.60. These descriptive measures provided an overview of the population characteristics before conducting further risk-factor and predictive analyses.

Table 3. Comparison of BMI and Age Category by Diabetes Status

Variable	No Diabetes	Diabetes
Mean BMI	28.10	31.96
Mean Age Category	7.85	9.38

Individuals with diabetes had higher average BMI and age-category values than those without diabetes. These differences suggest an association between diabetes status, BMI and age category in the dataset.

5.4 Correlation and Risk-Factor Analysis

Correlation analysis was performed using Spark's VectorAssembler and Pearson correlation function to examine the relationships between diabetes status (Diabetes-binary) and the selected health, demographic and lifestyle variables.The correlation coefficients were calculated and ranked according to their strength of association with diabetes status. Table 4 presents the variables with the strongest positive and negative correlations.

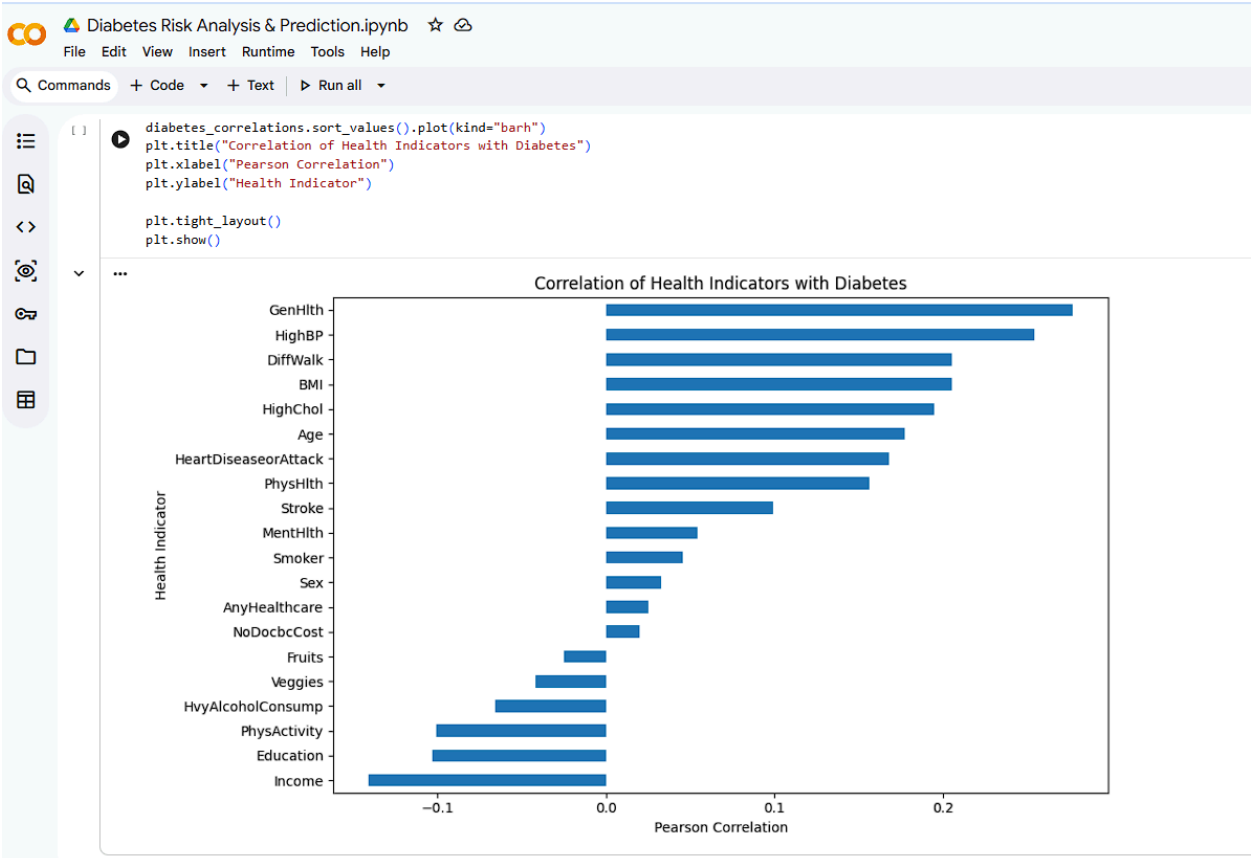
Table 4. Pearson Correlation Between Selected Variables and Diabetes Status

Variable	Correlation Coefficient
General Health	0.277
High Blood Pressure	0.254
Difficulty Walking	0.205
BMI	0.205
High Cholesterol	0.195
Age	0.177
Heart Disease/Heart Attack	0.168
Income	-0.141
Education	-0.103
Physical Activity	-0.100



```
#Correlation with diabetes
diabetes_correlations = (corr_pd["Diabetes_binary"].drop("Diabetes_binary").sort_values(key=abs, ascending=False))
print(diabetes_correlations)
```

```
GenHlth                0.276940
HighBP                 0.254318
DiffWalk               0.205302
BMI                    0.205086
HighChol               0.194944
Age                    0.177263
HeartDiseaseorAttack   0.168213
PhysHlth               0.156211
Income                -0.140659
Education             -0.102686
PhysActivity          -0.100404
Stroke                 0.099193
HvyAlcoholConsump     -0.065950
MentHlth               0.054153
Smoker                 0.045504
Veggies               -0.041734
Sex                    0.032724
AnyHealthcare          0.025331
Fruits                -0.024805
NoDocbcCost            0.020048
Name: Diabetes_binary, dtype: float64
```



The results show that general health and high blood pressure had the strongest positive correlations with diabetes status. BMI, difficulty walking and high cholesterol also showed positive associations. Income, education and physical activity had negative correlations.These findings suggest that health conditions and lifestyle characteristics are associated with diabetes status in the dataset. However, correlation does not establish causation.

5.5 Logistic Regression Model Development

A logistic regression model was developed using Apache Spark MLlib to predict diabetes status based on demographic, lifestyle and health-related variables. The dataset was divided into training (80%) and testing (20%) sets using a random split with seed=42.A Spark ML Pipeline was created using three main components:

1. VectorAssembler: Combines the 21 predictor variables into a single feature vector.
2. StandardScaler: Standardizes the feature values to support model training.
3. LogisticRegression: Trains the classification model using maxIter=50 and regParam=0.01.

The pipeline was fitted to the training data using pipeline.fit(train_df). The trained model was then applied to the testing data using model.transform(test_df) to generate predictions and

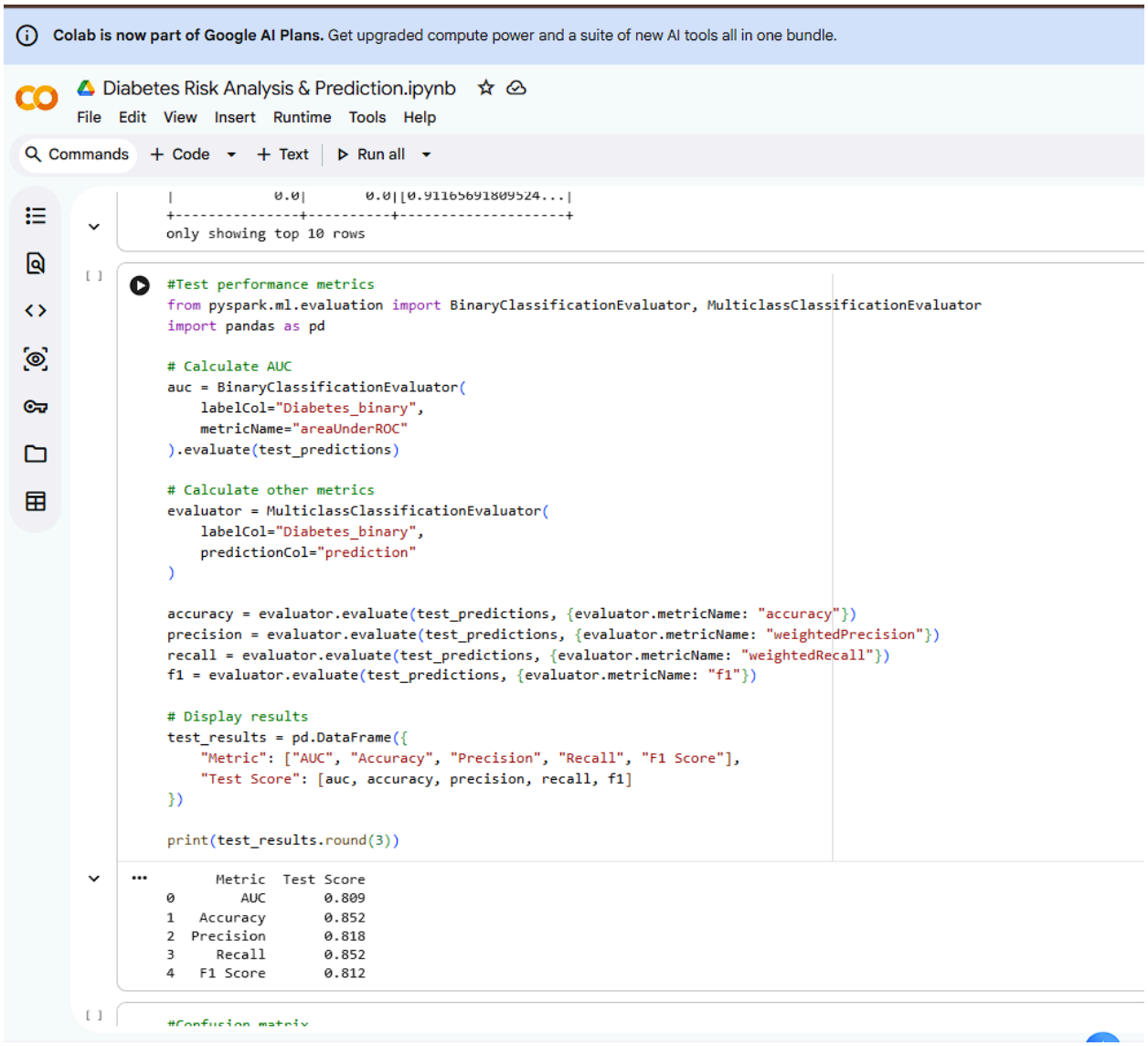
probability scores.Logistic regression was selected because the target variable, Diabetes-binary, has two possible outcomes: diabetes (1) and no diabetes (0).

5.6 Model Evaluation

The logistic regression model was evaluated using AUC, accuracy, precision, recall, and F1-score. A confusion matrix and ROC curve were also used to assess its classification performance.

Table 5. Model Performance Results

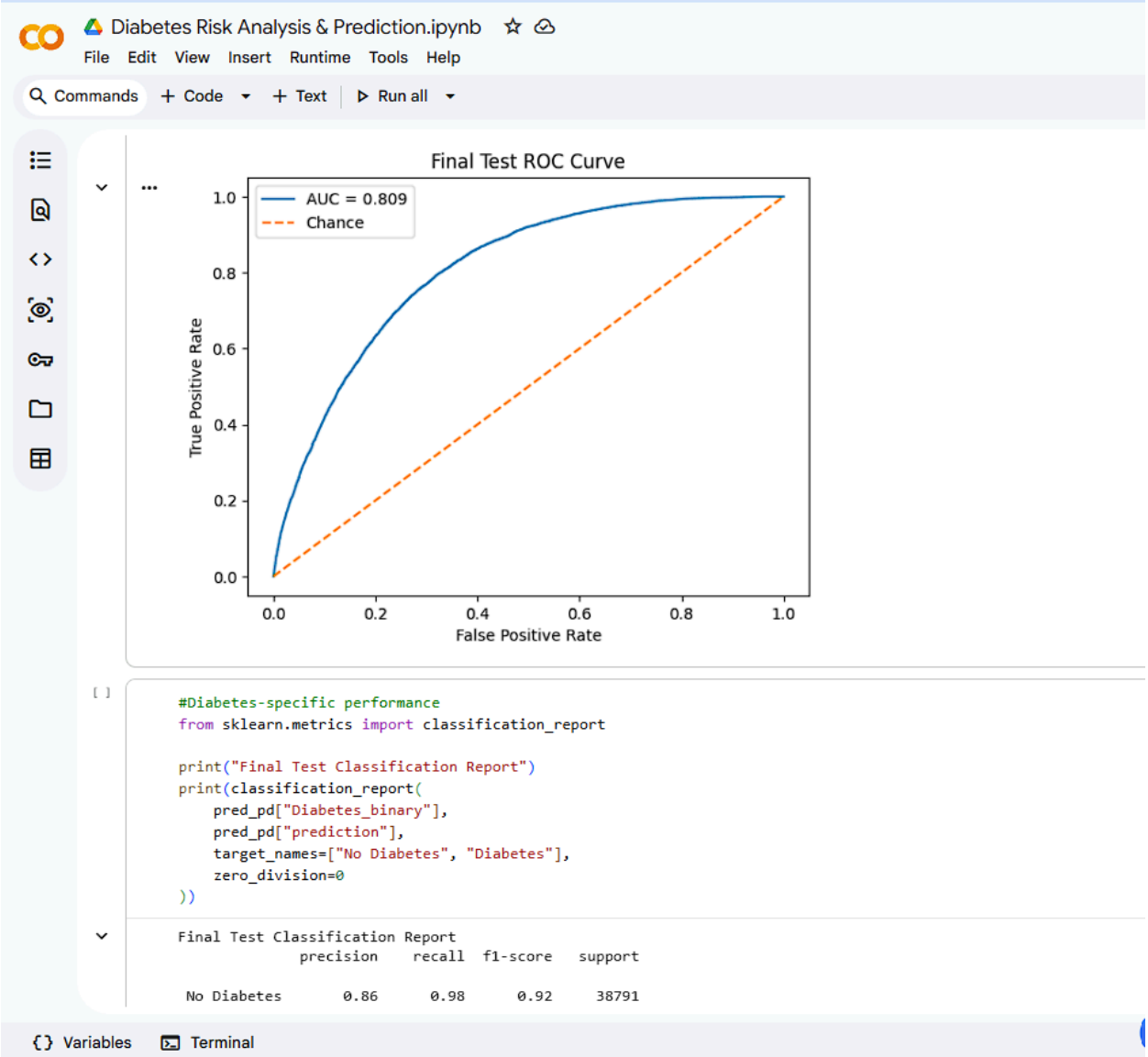
Evaluation Metric	Result
Area Under the Curve (AUC)	0.809
Accuracy	0.852
Precision	0.818
Recall	0.852
F1-Score	0.812



The model achieved an AUC of 0.809 and an accuracy of 85.2%, indicating that it was able to distinguish between the two outcome categories to a useful degree. However, accuracy alone does not fully describe its performance because the dataset contains more individuals without diabetes than individuals with diabetes.

Table 6. Classification Report by Diabetes Category

Diabetes Category	Precision	Recall	F1-Score	Support
No Diabetes	0.86	0.98	0.92	38.791
Diabetes	0.56	0.14	0.22	6.951



The model performed better at identifying individuals without diabetes than those with diabetes. Its recall for the diabetes category was only 0.14, meaning it identified approximately 14% of actual diabetes cases in the test data.The confusion matrix recorded 38,051 true negatives, 740 false positives, 6,008 false negatives, and 943 true positives. These results indicate that the model missed a substantial number of individuals with diabetes.Overall, although the model achieved reasonable AUC and accuracy, its low recall for diabetes cases highlights the need for further improvement before it could be considered for practical screening applications.

6.Conclusion

This project used Apache Spark to analyze the CDC Diabetes Health Indicators dataset and develop a logistic regression model for predicting diabetes status. The analysis identified several health-related and lifestyle characteristics associated with diabetes, with general health, high blood pressure, BMI, and difficulty walking showing notable positive correlations.The logistic regression model achieved an AUC of 0.809 and an accuracy of 85.2%. However, its low recall for individuals with diabetes (14%) highlighted the challenges of predicting outcomes in an imbalanced dataset. This demonstrates the importance of considering multiple evaluation metrics rather than relying on accuracy alone.Overall, the project demonstrated how big data analytics and machine learning can be applied to population health data to identify disease-related patterns and support predictive analysis. Future improvements could include addressing class imbalance, testing other machine-learning algorithms and adjusting classification thresholds to improve the identification of individuals with diabetes.

7.References

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